

Functions and Operations

UChicago Political Science Math Camp

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Today's Challenge

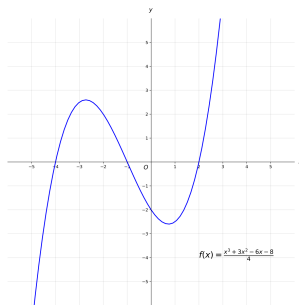
- ▶ You likely already know what a function is, or at least how to use one.
- ▶ We are going to *formally redefine* what a function is and how we might expect to see them.
- ▶ Focus is not just on the outcome, but on the **process** and **definitions**.

What you've seen already

You've likely seen a function expressed as y or $f(x)$, like

$$y = mx + b \qquad f(x) = ax^2 + bx + c$$

You also likely understand a graphed function, like



What is a function?

A function is a **mapping** of elements in some set (the **domain**) to elements in another set (the **codomain**):

$$f : A \rightarrow B$$

Such that

- ▶ f labels the function
- ▶ A is the domain (a set)
- ▶ B is the codomain (also a set)
- ▶ $f(x \in A) \rightarrow f(x) \in B$

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A function is a rule transforming **inputs** into **outputs**.

Let's start with an arbitrary function:

$$A = \{51, 52, 53, 54, 55\}$$

$$B = \{\text{Dorchester, Kenwood, Kimbark, Woodlawn, University}\}$$

Some function $g : A \rightarrow B$ could lend:

$$g(51) = \text{Dorchester}$$

$$g(52) = \text{Kimbark}$$

$$g(53) = \text{Woodlawn}$$

$$g(54) = \text{University}$$

$$g(55) = \text{Kenwood}$$

Why? Because we said so!

Functions in Political Science

We usually use a function to take some **variable**(s) as an input and return some **value** as an output.

- ▶ A *variable* is a symbol/letter that represents some unknown value

For example, we might take:

$$Wealth : \{\text{Resources, Institutions, FDI, IR}\} \rightarrow GDP$$

What might these parent sets look like?

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What other functions in political science might we create?

A Specific Social Science Function

How can we calculate GDP per capita from known, well-defined inputs?

- ▶ Through function notation $f : A \rightarrow B$
- ▶ Through some mathematical rule defining f and returning $f(A) \in B$

In reality, we usually (but not always) work with more interpretable functions. Let us define:

$$C = \{x \in \mathbb{N} : x < 5\} \qquad D = \{2 \cdot x \in \mathbb{N} : x < 5\}$$

Let us construct some function $h : C \rightarrow D$ such that $h(c) = 2c$:

$$C = \left\{ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} \right\} \rightarrow h(c) = 2c \rightarrow \left\{ \begin{matrix} h(1) = 2 \\ h(2) = 4 \\ h(3) = 6 \\ h(4) = 8 \end{matrix} \right\} = D$$

So, h is a very clean mapping from C to D . But, this is a very limiting set of assumptions.

What if we're not so lucky?

Most often, we are using functions to map onto some real space $\mathbb{R}^n$.
Let's redefine:

$$h : C \rightarrow \mathbb{R}$$

$$C = \begin{Bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{Bmatrix} \rightarrow h(c) = 2c \rightarrow \begin{Bmatrix} h(1) = 2 \\ h(2) = 4 \\ h(3) = 6 \\ h(4) = 8 \end{Bmatrix} \in \mathbb{R}$$

Now, we can broaden our codomain while preserving the nature of the function. But, there is a problem with this...

Bijection

A function is **bijection** iff it is both **injective** and **surjective**:

- ▶ A function $f : A \rightarrow B$ is injective iff f maps every element in A onto one and ONLY one element in B . (No two inputs return the same output.)

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- ▶ A function $f : A \rightarrow B$ is surjective iff every element in B is mapped by f from an element in A .

Which condition(s) does $h : C \rightarrow \mathbb{R}$ not satisfy?

Your Turn

Explain whether or not the following functions are bijective, and why:

► $f : \text{English letters} \rightarrow \{x \in \mathbb{N} : x \leq 26\}$

Where f returns for each letter its alphabetical order

► $g : \text{Animals in the zoo} \rightarrow \mathbb{R}_+$

where g returns for each animal its number of legs

► $h : \mathbb{Q}_+ \rightarrow \mathbb{R}_+$

where h returns a given arithmetic operation to its input

Inverse Functions

- ▶ A function's inverse returns its domain element (input) given its codomain element (output).
- ▶ We express an inverse domain as f^{-1} , such that:

$$f^{-1} : B \rightarrow A$$

$$h(2) = 4 \quad \Rightarrow \quad h^{-1}(4) = 2$$

- ▶ This is useful for when we know what output we have (or want), but want to find an input to produce it.

How do we invert a non-bijective function?

An Oasis

- ▶ This is a formally rigorous way to define *what you already know to be functions*: most “functions” in algebra and calculus just map $\mathbb{R} \rightarrow \mathbb{R}$
- ▶ Take $f(x) = x^2$:
 - ▶ It is implied that $x \in \mathbb{R}$, and we can see that $x \in \mathbb{R} \Rightarrow x^2 \in \mathbb{R}$

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 - ▶ Therefore, we have an input, an output, and a rule – all well-behaved.
 - ▶ ... right, guys?

Turns out $y = x^2$ is bullshit

- ▶ The function as defined is **not injective**: $x, -x$ take on the same value of $f(x)$.

Turns out $y = x^2$ is bullshit

- ▶ The function as defined is **not injective**: $x, -x$ take on the same value of $f(x)$.
- ▶ The function as defined is **not surjective**: it is not true that for every output $x \in \mathbb{R}$, we can recover $f^{-1}(x) \in \mathbb{R}$.
 - ▶ We accommodate for this by implicitly assuming that $f(x) : \mathbb{R} \rightarrow \mathbb{R}_+$, but not rigorous if not specified...

We don't need bijection to work with functions, but it can be quite useful in modeling and proofs, and in considering probabilistic distributions.

Dealing with $y = x^2$ anyways

- ▶ Suppose we are working with $f(x) = x^2$, and ignore the “bullshit” part
- ▶ Let's choose a few inputs; call that set X

$$X = \left\{ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} \right\} \rightarrow f(x \in X) = x^2 \rightarrow \left\{ \begin{matrix} f(1) = 1 \\ f(2) = 4 \\ f(3) = 9 \\ f(4) = 16 \end{matrix} \right\} \in \mathbb{R}$$

- ▶ Now, let's plot these results.
- ▶ What happens as X begins to resemble $\mathbb{Q}$? What about $\mathbb{R}$?

Desmos

A useful tool for visualizing functions

This is also a reminder for me to open it now

Common Functions: Linear

- ▶ A constant relationship between x and $f(x)$. Often expressed as

$$f(x) = mx + b$$

where m is the rate of change of $f(x)$ over a unit in x , and $b = f(0)$.

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- ▶ Or, if you're feeling adventurous,

$$\exists a \in \mathbb{R} \text{ s.t. } \frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}} = a \quad \forall i \in \mathbb{Z}$$

Creating Linear Functions

Assuming we haven't been given m and b , we then:

- ▶ Calculate $b = f(0)$ and

$$m = \frac{\Delta f(x)}{\Delta x}$$

Then plug in for $f(x) = mx + b$

- ▶ Or, for some given m and pair $x, f(x) = (x_i, f(x_i))$, use **point-slope form**:

$$m(x - x_i) = f(x) - f(x_i)$$

Common Functions: Polynomial

- ▶ An algebraic function where inputs are multiplied by themselves some whole number of times over
- ▶ The highest exponent determines the **order** of the polynomial
- ▶ For example, a quadratic function might look like this:

$$f(x) = ax^2 + bx + c$$

and is a **second order polynomial**.

Roots & Wrangling Polynomials

There's no uniform way to deal with polynomials, but there are a few common traits:

- ▶ We often want to solve a polynomial for $f(x) = 0$. We call the x value(s) that produce $f(x) = 0$ the **root(s)**.
- ▶ (Even if we want to solve $f(x) = a$, we will usually express $f(x) - a = 0$ and absorb $-a$ into $f(x)$.)

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 - ▶ (Even if we want to solve $f(x) = a$, we will usually express $f(x) - a = 0$ and absorb $-a$ into $f(x)$.)
- ▶ The **most** roots a polynomial can have is equivalent to its order.
- ▶ We include $\sqrt{}$ functions as polynomials, where:

$$x^{\frac{a}{b}} = \sqrt[b]{x^a}$$

Common Functions: Exponential

- ▶ An exponential function consists of some constant being multiplied by itself a number of times dependent on x :

$$f(x) = a^x$$

- ▶ A special case of the exponential function is when $a = e$, such that

$$f(x) = e^x$$

Which is sometimes abbreviated as

$$f(x) = \exp(x)$$

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 - ▶ Any non-integer x 's then call for a root of a negative number, which is only real for odd integers (contradiction)

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 - ▶ Any non-integer x 's then call for a root of a negative number, which is only real for odd integers (contradiction)
- ▶ Many (most?) common asymmetric probability distributions are exponential (i.e., low values happen quite frequently while high values happen infrequently)

Factorials!

Factorials are a very special family of functions with the following property:

$$n! = 1 \times 2 \times 3 \times \dots \times (n-1) \times n$$

Note that factorials are discrete, and then only well-defined for $n \in \mathbb{N}$. The Γ function allows us to represent factorials for $x \in \mathbb{R}, x \notin \mathbb{N}$.

We define $0! = 1$.

Wrangling Factorials

Factorials are rapidly-increasing, and only interpretable for small n inputs (i.e. $10! = 3,628,800$). But, we often use some nifty division:

$$\frac{(n+1)!}{n!} = \frac{1 \times 2 \times \dots \times (n-1) \times n \times (n+1)}{1 \times 2 \times \dots \times (n-1) \times n}$$

We can't easily calculate¹ $15!$ or $13!$, but it's very easy to calculate $\frac{15!}{13!}$

¹by hand – computers can go a bit higher, Google goes up to $170! = 7.25 \times 10^{306}$

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The goal with factorials is to obtain a quotient!

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Logarithms

- ▶ Logarithms are the inverse of exponential functions, and help identify what exponent an input must be raised to in order to produce a given output

$$x = b^y \quad \Rightarrow \quad \log_b x = y$$

- ▶ Though often not intuitive in calculation, can be very useful in regressing percentages or representing widely-distributed data
- ▶ (Formal theorists also like logarithms as they are monotonic and bijective W.R.T. the positive reals.)
- ▶ Temperature vs. Decibels

Wrangling Logarithms

- ▶ You may be used to assuming $b = 10$ if unspecified, such that

$$\log 100 = 2$$

This is commonly used for data analytics and empirical measures.

- ▶ But, in advanced mathematics (and formal theory), we sometimes assume that $b = e$, represented as

$$\ln x = n \quad \Rightarrow \quad e^x = n$$

sometimes even if written as log and not as ln.

Multivariable Functions

- ▶ It's not all that often that we're only working with a single input and output. For example, projected GDP is a function of both growth rate and present GDP.
- ▶ We can represent a multivariable function as:

$$f(x, y), f(x, y, z), f(w, x, y, z), \dots$$

Where f returns a single output given each of these inputs.

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Where f returns a single output given each of these inputs.

- ▶ Informally, we can think of the output of $f(x)$ as a line, and $f(x, y)$ as a surface, but difficult to conceptualize beyond that
- ▶ We will learn how to wrangle these in the calculus unit (it's easier than it sounds)

Mapping Between Different Dimensions

- ▶ Recall that our “conventional” single-variable function maps $f : \mathbb{R} \rightarrow \mathbb{R}$
- ▶ When we have multiple variables, though, we are mapping from a “space” of those variables. So, it might be the case that for $f(x, y)$,

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}$$

$$f^{-1} : \mathbb{R} \rightarrow \mathbb{R}^2$$

- ▶ This can be much harder, though... to apply an inverse function, we often want only one space

Hints for Multivariable Functions

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Hints for Multivariable Functions

- ▶ Hint #1: use $\mathbb{R}^n$ for n input variables
- ▶ Hint #2: we are often just mapping something onto $\mathbb{R}$, which means that for the input(s), *we just recover a single numeric output.*
- ▶ Hint #3: don't try to visualize functions involving beyond $\mathbb{R}^3$ (and really not beyond $\mathbb{R}^2$)

A Few Helpful Notes

- ▶ The **support** of a function is the set of reals which returns a non-zero output
- ▶ It is almost always helpful to use **Desmos** to plot a function that you cannot visualize
- ▶ Functions can be “summed”: it can be the case that one term of f is polynomial and another term is exponential, etc.

Taking A Sum

- ▶ We often want to calculate the total amount of individual values:
 - ▶ Total wealth is the sum of individual wealth for every member of the population
 - ▶ Global emissions are the sum of national emissions for every nation in the world
 - ▶ Total contributions is the sum of all individual contributions for every contributor
- ▶ Or, even if not substantively, we just want to use simpler notation

Summation Operation

To take a sum, we use the summation operation:

$$X = \sum_{i=a}^b x_i$$

Where

- ▶ X is the total sum
- ▶ i is the **index** (indexes must be natural numbers)
- ▶ a is the **lower bound**
- ▶ b is the **upper bound**

Some Examples

We can express $1 + 2$ as

$$\sum_{i=1}^2 i = 3$$

And even use set-builder notation

$$\sum_{i=1}^2 x_i : x \in \mathbb{N} = 3$$

We can get more complicated, but the concept stays the same:

$$f(x) = \sum_{i=1}^N \frac{\lambda}{\sqrt{i}} = \frac{\lambda}{\sqrt{1}} + \frac{\lambda}{\sqrt{2}} + \dots + \frac{\lambda}{\sqrt{N-1}} + \frac{\lambda}{N}$$

More Summation

- Sometimes our domain may even be the bounds:

$$f(x) = \sum_{i=1}^x \frac{1}{2^i}$$

- But, note that we must specify where we are starting from (often, but not always, either $i = 0$ or $i = 1$)
- Any symbol can be used as an index, but we most often use either i, k, n

Wrangling Summations

- ▶ A summation expression acts like a distributive term, and can be treated as such:

$$\sum_{i=1}^N a \cdot f(i) = a \cdot \sum_{i=1}^N f(i)$$

but

$$\sum_{i=1}^N (f(i))^b \neq \left(\sum_{i=1}^N f(i) \right)^b$$

Products

A nearly-identical process can be used to multiply, rather than add, a series of indexed values:

$$X = \prod_{i=a}^b x_i$$

Where

- ▶ X is the total product
- ▶ i is the **index** (indexes must be natural numbers)
- ▶ a is the **lower bound**
- ▶ b is the **upper bound**

Sequences

Sequences are special cases of functions:

- ▶ A **sequence** is a function $a : \mathbb{N} \rightarrow \mathbb{R}$

Specifically, sequences have *discrete inputs* and *indexed outputs*

Series

Series are sums of infinitely many numbers:

$$\sum_{k=1}^{\infty} \frac{1}{2^k}$$

$$\Rightarrow \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$$

How might we define a series in relation to a sequence?

A Very Important Series

$$\sum_{k=0}^{\infty} \frac{1}{n!} = e$$